

Package #112

Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'appropriate_device_problem_term_code_not' for shoulder prosthesis: 23 MAUDE reports across 3 distinct devices (20200506 to 20211104); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: appropriate, available, conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurrin

Mechanism: Passive structural optimization with topology optimization

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.